

Biology
BIOLOGY 1A02
Foundations and Applications of
Cellular and Molecular Biology
Winter 2027



SCIENCE

Instructor Information

Dr. Lovaye Kajiura
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Office Hours:
TBA

Course Information

This document represents a draft course outline that is posted early to provide students with as much information as possible in advance of registration; information was derived from the most recent offering of the course. The final outline might contain different details and will be provided at the start of the course.

IMPORTANT: To complete the 3-credit equivalency of BIO1A03, you must enroll in BOTH BIO1A02 and BIO1A01. BIOLOGY 1A02 is the theory/concepts/applied knowledge portion of the full, 3-credit BIOLOGY 1A03 course.

This course is an online/asynchronous learning format, with three tests that are written "in-person" on McMaster University's campus and are invigilated by our instructional team. All lectures will be provided as pre-recorded online modules on Avenue to Learn. Pedagogically, this resource will also facilitate the ability for students to self-pace their learning and revisit and review lectures more than once as a study tool. The module availability will also enable inclusive access for students who need closed captioning of course lectures. We encourage all students to follow the schedule as posted on Avenue, to ensure that they do not fall behind in course progression, and are best prepared for all assessments.

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Course Dates: 01/04/2027 - 04/07/2027

Units: 2.00

Course Delivery Mode: Online

Course Description: Structure, molecular composition and function in sub-cellular and cellular systems. Three hours (virtual lectures, web modules); one term Cross-list(s): BIOLOGY 1A03 Prerequisite(s): One of Grade 12 Biology U, BIOLOGY 1P03 or registration in an Engineering program Antirequisite(s): BIOLOGY 1A03; Not open to students with credit or registration in HTHSCI 1I06 A/B or ISCI 1A24 A/B. Students may take BIOLOGY 1A03 (or BIOLOGY 1A01 and 1A02) and 1M03 in any order but are strongly encouraged to take them in different terms.

Instructor-Specific Course Information

BIO1A02 teaches foundational molecular and cellular biology according to the 5 Themes below:

Course Themes
Course Orientation & Introduction Lecture
Theme 1: The Structure of the Cell
Theme 2: From Gene to Protein
Theme 3: Responding to the Environment
Theme 4: DNA Replication & Mitosis
Theme 5: The Principles of Inheritance

Academic Property Statement

No part of the BIOLOGY 1A02 lectures, modules, course information, practice tests, tests, or test answer keys, may be reproduced or posted, in any form or by any means, without permission in writing by professor Dr. Lovaye Kajiura. No visual media, voice recordings, PowerPoint slides, PDFs, MP3 or MP4 media, or

lecture-related information may be reproduced or communicated by any means. Copying BIOLOGY 1A02 lecture or laboratory materials for distribution (e.g. uploading materials to a commercial website) is strictly prohibited.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Microsoft Teams (virtual office hours and discussion board):

<https://office365.mcmaster.ca/office-365-3/applications/>

Microsoft Office: <https://office365.mcmaster.ca/office-365-3/applications/>

Course Learning Outcomes

- The BIOLOGY 1A02 course is designed for students who intend to specialize in Science programs and is required for many upper level courses in the Faculty of Science. Upon completion of BIOLOGY 1A02, students will be able to: 1. Effectively discuss the fundamental concepts and underlying processes related to cellular and molecular biology and 2. Apply these concepts to real-world innovations and biomedical applications.
- The primary goal of the course is to prepare students academically for subsequent, specialized Biology courses and to ensure that students acquire skills essential for upper-level biology courses and biology related fields of study.

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

**Textbook - Biology: How Life Works****ISBN:** ISBN: 9781319562298 (e-book)**Authors:** James Morris; Daniel Hartl; Andrew Knoll; Robert Lue; Melissa Michael; Andrew Berry; Andrew Biewener; Brian Farrell; N. Michele Holbrook; Jessica Liu; Jean Heitz; Mark Hens; Elena Lozovsky; John Merrill; Randall Phillis; Debra Pires**Publisher:** Macmillan Learning**Publication Date:** 2023**Edition:** Fourth Edition

Assigned Textbook Readings - Refer to the Bio1A02 Avenue course website for the assigned reading list.

Macmillan Achieve Resource for Students - compliments both the e-book and loose-leaf textbook options. Achieve will be used to facilitate learning and not graded.

Calculator

Only McMaster approved calculators (Casio fx-991MS or Casio fx-991MS Plus) will be permitted for use during the tests.

Class Format

Online

LECTURES

BIOLOGY 1A02 lectures are taught in an online/asynchronous lecture format using the online Learning Management System, Avenue to Learn. Please note that the Biology 1A02 tests will be written "in-person" on McMaster University's campus. For lectures, this means that you will learn the course content at your own pace and on your own time using online visually rich web modules. The online modules will be released weekly. The following week, an online review and applied lecture will be provided. The review lecture will give you the opportunity to revisit concepts from the core online modules that you may have found challenging. Additionally, the applied lectures will explore applied contexts with examples from innovative research and data analytics.

The BIOLOGY 1A02 review and applied lectures will be a synthesis of several sources of information. Please note that students are expected to study all lecture

materials. Students will benefit the most by including their own detailed written or typed notes into the provided BIOLOGY 1A02 lecture resources and slides posted on Avenue.

In addition to the online discussion board, weekly live online office hours will be hosted by Dr. Kajiura on Microsoft Teams. This will provide all students the opportunity to interact directly with the course instructor and ask questions related to the online modules, review and applied lecture content.

How to log into the BIOLOGY 1A02 Avenue Site

1. Start your web browser and go to: <http://avenue.mcmaster.ca>
2. Click on the Login button.
3. Enter your McMaster email address and corresponding password that you use to sign into your email and other Office 365 applications.
4. Locate the BIOLOGY 1A02 course from your Avenue homepage, typically below the general Avenue announcements or from the drop-down menu at the top.

Microsoft Teams

Microsoft Teams will be used in BIOLOGY 1A02 to facilitate learning by providing opportunities for live virtual interaction with the course instructor during their office hours. In addition, students will also be able to engage in discussion of course content with the instructor and other students enrolled in the course by posting questions in the appropriate channels. Students will be able to download and install the Microsoft Teams app from their McMaster Office 365 account to access the BIOLOGY 1A02 course and lab section-specific Microsoft Teams channels. Instructions regarding how to use Microsoft Teams and support are found on the McMaster University Technology Services website:

<https://office365.mcmaster.ca/>

Course Technology Requirements

To follow and participate in this online class format it is expected that you have reliable access to the following:

- A computer that meets performance requirements found here.
- An internet connection that is fast enough to stream video.

- Computer accessories that enable class participation, such as a speakers and webcam when needed.

If you think that you will not be able to meet these requirements, please contact uts@mcmaster.ca as soon as you can. Please visit the Technology Resources for Students page for detailed requirements. If you use assistive technology or believe that our platforms might be a barrier to participating, please contact Student Accessibility Services, sas@mcmaster.ca, for support.

Course Evaluation

Final BIOLOGY 1A02 grades will be determined by the following grading scheme:

TERM TEST 1 - Monday, February 1st, 7:00pm-8:30pm (90 minutes). Test 1 will be held in-person on McMaster University's campus. Test 1 will consist of 30 multiple-choice questions and 10 marks worth of written answer questions, covering Theme 1 (Modules 1, 2, 3, 4), & Theme 2 (Modules 1,2) as well as the Introductory lecture, the related Review lectures, Applied lectures, and supplementary resources related to Theme 1 Modules 1,2,3,4 & Theme 2 Modules 1, 2. **Test 1 is worth 30% of the final grade.**

TERM TEST 2 - Monday, March 1st, 7:00pm-8:30pm (90 minutes). Test 2 will be held in-person on McMaster University's campus. Test 2 will consist of 30 multiple-choice questions and 10 marks worth of written answer questions, covering Theme 2 (Modules 3,4), Theme 3 (Modules 1,2,3,4), and the related Review lectures, Applied lectures, and supplementary resources related to Theme 2 Modules 1,2, Theme 3 Modules 1,2,3,4. **Test 2 is worth 35 % of the final grade.**

MAKEUP TESTS will be held on **Monday March 22nd** 7:00pm, 90 minutes, location to be announced.

TERM TEST 3 - Tuesday, March 30th, 7:00pm-8:30pm (90 minutes). Test 3 will be held in-person on McMaster University's campus. Test 3 will consist of 30 multiple-choice questions and 10 marks worth of written answer questions, covering Theme 4 (Modules 1,2,3,4), Theme 5 (Modules 1,2,3,4) , as well as the related Review

lectures, Applied lectures, and supplementary resources related to Theme 4 Modules 1,2,3,4, and Theme 5 Modules 1,2,3,4. **Test 3 is worth 35 % of the final grade.**

TOTAL= Test 1 (30%) + Test 2 (35%) + Test 3 (35%) = 100%

BIO1A02 Test Format - Assessments may include multiple-choice questions, questions with figures or graphs, and written factual, conceptual, and application-style questions. The BIOLOGY 1A02 tests will be completed individually and are NOT open-book. To best prepare students for each test, a practice test will be made available on Avenue well in advance of each test. While the practice tests are not worth any portion of your final grade in the course, students will benefit by becoming more comfortable with the testing format and reviewing the test content.

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

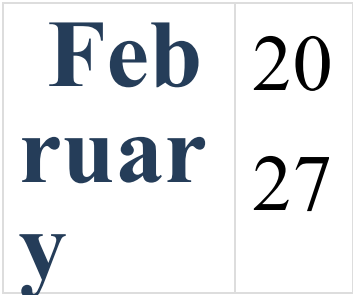
Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

A detailed course schedule (calendar of events) will be posted at the start of the Winter academic term.

Jan 20 uary 27						
MON	TUE	WED	THU	FRI	SAT	SUN
				1	2	3
4	5	6	7	8	9	10
	Intro Lecture	Applied Lecture	THEME 1 Modules 1,2 Released			
11	12	13	14	15	16	17
	THEME 1 Modules 1,2 Review Lecture	THEME 1 Modules 1,2 Applied Lecture	THEME 1 Modules 3,4 Released			
18	19	20	21	22	23	24
	THEME 1 Modules 3, 4 Review Lecture	THEME 1 Modules 3,4 Applied Lecture	THEME 2 Modules 1,2 Released			
25	26	27	28	29	30	31

THEM	THEM	THEM
E 2	E 2	E 2
Modul	Modul	Modul
es 1,2	es 1,2	es 3,4
Review	Applic	Releas
Lectur	d	ed
e	Lectur	
	e	



MON	TUE	WED	THU	FRI	SAT	SUN
1	2	3	4	5	6	7
TES T #1	THEM E 2	THEM E 2	THEM E 3			
	Modul es 3,4	Modul es 3,4	Modul es 1,2			
	Review	Applic	Releas			
	Lectur e	d	ed			
		Lectur e				
8	9	10	11	12	13	14

	THEM E 3 Modul es 1,2 Review Lectur e	THEM E 3 Modul es 1,2 Applic d Lectur e	THEM E 3 Modul es 3,4 Releas ed		T	
15	16	17	18	19	20	21
	MID- TERM	RECE SS				
22	23	24	25	26	27	28
	THEM E 3 Modul es 3,4 Review Lectur e	THEM E 3 Modul es 3,4 Applic d Lectur e	THEM E 4 Modul es 1,2 Releas ed			

Mar ch	2027					
MON	TUE	WED	THU	FRI	SAT	SUN
1	2	3	4	5	6	7

TEST #2	THEM	THEM	THEM			
	E 4	E 4	E 4			
	Modul	Modul	Modul			
	es 1,2	es 1,2	es 3,4			
	Review	Applic	Releas			
	Lectur	d	ed			
	e	Lectur				
		e				
8	9	10	11	12	13	14
	THEM	THEM	THEM			
	E 4	E 4	E 5			
	Modul	Modul	Modul			
	es 3,4	es 3,4	es 1,2			
	Review	Applic	Releas			
	Lectur	d	ed			
	e	Lectur				
		e				
15	16	17	18	19	20	21
	THEM	THEM	THEM			
	E 5	E 5	E 5			
	Modul	Modul	Modul			
	es 1,2	es 1,2	es 3,4			
	Review	Applic	Releas			
	Lectur	d	ed			
	e	Lectur				
		e				
22	23	24	25	26	27	28
MAK	THEM	THEM				
EUP	E 5	E 5				
TEST	Modul	Modul				
#1	es 3,4	es 3,4				
TEST	Review	Applic				
#2	Lectur	d				
	e	Lectur				
		e				
29	30	31				

TES
T #3

Apri
l

20
27

MON	TUE	WED	THU	FRI	SAT	SUN
			1	2	3	4
5	6	7	8	9	10	11
				EXAM S BEGIN		
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI: Some Use Permitted

Students are prohibited from using generative AI to submit all graded components. In alignment with McMaster academic integrity policy, it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also state in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Students may only use generative AI in this course to facilitate learning.

Course Specific Relief for Missed Academic Work Information

BIOLOGY 1A02 MSAF Policy Table

Missed academic work must be accompanied by a McMaster Student Absence Form (MSAF). Please follow the instructions below:

1. For academic work worth less than 25%, MSAF must be requested online through Mosaic.
2. For academic work worth more than 25%, MSAF must be requested by emailing your Faculty's office of the associate dean.
3. Wait for MSAF to be approved
4. Follow up with the professor, Dr. Kajiura, regarding your MSAF

BIOLOGY 1A02 MSAF POLICY TABLE

Missed Academic Work	Result After MSAF is Submitted and Approved
Missed Test 1	A Make-Up Test 1 will be provided to students (see calendar). Students who do not submit an approved MSAF or fail to write the make-up test will receive a grade of 0% for the test.
Missed Test 2	A Make-Up Test 2 will be provided to students (see calendar). Students who do not submit an approved MSAF or fail to write the make-up test will receive a grade of 0% for the test.
Missed Test 3	There is no makeup test for Test 3. To defer this test, you must contact your Faculty's Associate Dean's office. If your deferral is approved, you will write Test 3 during the next scheduled deferred examination period. Students without an approved deferral will receive a grade of 0% for Test 3.

For more details on MSAFs, please visit the Policy on Requests for Relief for Missed Academic Term Work: <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/msaf-policy/>

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. All grade concerns and discrepancies must be reported to the professor within a week of receiving the grade.

Regrade Policy

Office hours with the professor will be arranged following assessments to address concerns students may have regarding their grade. Requests for re-evaluation of tests or assignments must be made in writing to the professor using the form provided in Avenue and justifying the reason for which the assignment should be remarked. This should be done within one week of discussing the marked test or assignment with the instructional team and BEFORE the final exam is written. If the professor considers the written justification to be insufficient (e.g. simply wanting a higher grade is insufficient), the assignment will not be re-graded. If the justification is considered sufficient, the entire assignment/test will be re-graded. Please be aware that this may result in an increase, decrease or no change to the original mark. There will be no alternative assignments that can be completed by students for the purpose of increasing their final grade.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when

they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students

should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.